

0829.20

5305

**GERMINATION OF THE HETEROCYST IN
TWO MEMBERS OF THE RIVULARIACEÆ,
GLOEOTRICHIA RACIBORSKII WOLOSZ.
AND *RIVULARIA MANGINI* FREMY***

By T. V. DESIKACHARY, M.Sc.

University Botany Laboratory, Madras

Received for publication on September 15, 1945

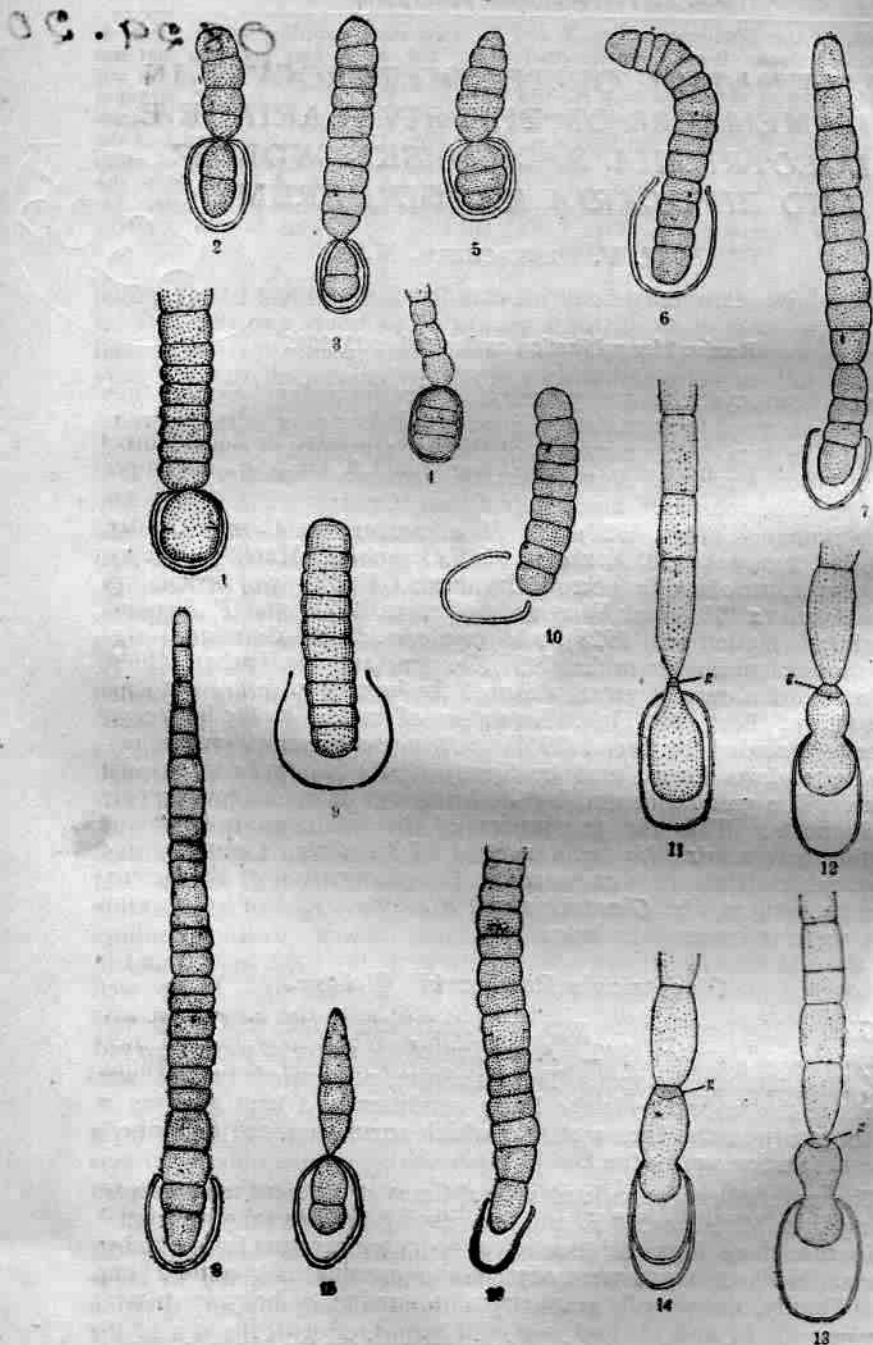
USUALLY heterocysts do not germinate. But in a few genera they have been observed to germinate. Even here it is not a very common or regular feature. Germination of the heterocyst has been recorded so far in four species of *Nostoc*, viz., *N. commune* Vaucher (Geitler, 1921), *N. ellipso sporum* (Desm.) Rabenhorst (Geitler, 1921), *N. Linckia* (Roth) Bornet (Geitler, 1921) and *N. microscopium* Carm. (Geitler, 1921), and two species of *Anabaena*, viz., *A. variabilis* Kützinger (Geitler, 1921) and *A. steloidea* Canabæus (Canabæus, 1929), two species of *Tolypothrix*, viz., *T. lanata* Wartmann (Geitler, 1921) and *T. Elenkinii* Hollerbach (Hollerbach, 1928), and one species of *Calothrix*, viz., *C. Weberi* Schmidle (Steinecke, 1931). Iyengar and Desikachary (1944) have recorded a case of germination of the heterocyst in *Brachytrichia Balani* (Lloyd) Born. and Flah. Formation of gonidia by the heterocyst has been recorded by Brand (1901) in *Nostoc commune* Vaucher and *N. microscopium* Carm. and by Spratt (1911) in *Anabaena cycadeæ* Reinke. The gonidia formed by the heterocyst later on develop into new filaments. Thus the germination of the heterocyst is known in only five genera, viz., *Nostoc*, *Anabaena*, *Tolypothrix*, *Calothrix* and *Brachytrichia*. The writer came across the germination of the heterocyst in two genera, viz., *Gloeotrichia* and *Rivularia*. A detailed account of the germination is given below :—

***Gloeotrichia Raciborskii* Wolosz.**

Gloeotrichia Raciborskii was collected from a lake in Chingleput, in the month of January 1942. A portion of the material was preserved in 4% Formalin on the spot. The remaining portion of the alga was brought to the laboratory in the living condition and kept growing in the laboratory in the lake water to which some sterilized Schreiber's nutrient solution was added.

The material was vernalized several times at different temperatures ranging from 32° F. to 50° F. and for varying periods ranging from a few hours to three days, and then left at room temperature for a number of days. No heterocysts were observed to germinate for quite a long time. Finally, in one of the attempts, a number of heterocysts were

* This paper formed part of a thesis approved for the Master of Science Degree of the Madras University.



Text-figs. 1-16.—Figs. 1-14. Stages in the germination of the heterocyst in *Gloeotrichia Raciborskii* Wolosz. Fig. 1. Beginning of the first division of the

contents of the heterocyst. Figs. 2 and 3. Two-celled germling stages. Fig. 4. Four-celled stage. Fig. 5. Three-celled stage. Fig. 6. A long germling that has grown out of the heterocyst. Figs. 7 and 8. Long germlings that have grown out of the heterocyst connected with the old trichome. Fig. 9. A ten-celled germling with traces of the old heterocyst wall still persisting at the base. Fig. 10. Germling fully come out of the old heterocyst wall. Figs. 11-13. Various stages of the coming out of the contents of the heterocyst without undergoing any division (g = granule). Fig. 14. Heterocyst rejuvenating for a second time. Note the two heterocyst walls. Figs. 15 and 16. Stages in the germination of the heterocyst in *Rivularia Mangini* Frey. Figs. 1, 2, 3, 10, 15, $\times 1150$; Figs. 5-9, 16, $\times 1100$; Figs. 11-14, $\times 1450$; Fig. 4, $\times 800$.

found to have germinated. In this case the material had been exposed to a temperature of 45° F. for a period of 24 hours and then left at room temperature. This material was kept under observation and all the details of germination up to the fully developed trichome were carefully followed.

The heterocyst in the alga is situated at the base of the filament, and has a single pore on the trichome side. A large granule is present very near the pore.

Germination of the heterocyst.—The contents divide into two cells (Text-figs. 2 and 3; Pl. I, Fig. 2). The beginning of the cross-wall formation can be seen in Text-fig. 1 and Pl. I, Fig. 1; and in Text-figs. 2 and 3 and in Pl. I, Fig. 2 the division into two cells is complete. The granule present near the pore of the heterocyst evidently disappears at the commencement of the division (Text-fig. 1; Pl. I, Fig. 1). By further division of both these cells a four-celled germling is formed (Text-fig. 4; Pl. I, Fig. 3). Sometimes only one of the two cells divides and then only three cells are seen in the germling (Text-fig. 5; Pl. I, Fig. 5). As the germling grows longer and emerges out through the pore, it is seen still connected with the rest of the trichome (Text-figs. 7 and 8; Pl. I, Fig. 4). Whether, after growing further, this germling will break away from the old trichome and become an independent trichome or will continue to grow attached to the old trichome could not be definitely stated from the stages of germination available in the material. But a few cases of well grown germlings with the old heterocyst wall still persistent at the base were found in the material (Text-figs. 6, 9 and 10; Pl. I, Fig. 6). These well grown germlings were quite independent and were not connected with the old trichome portions. These germlings might have broken away from the old trichome or might merely be the product of the germination of isolated heterocysts.

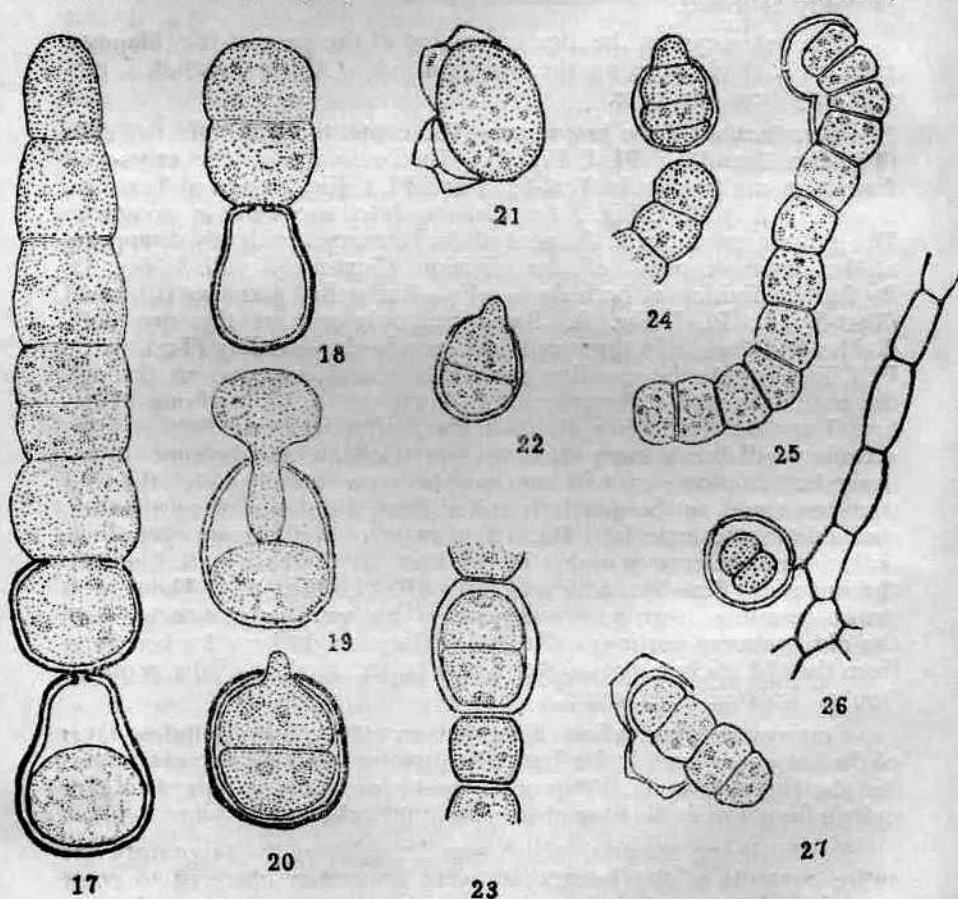
During the germination of the heterocyst the inner cellulose layer of the heterocyst wall is the first to be dissolved and the outer layer gets dissolved away later on. This outer layer is generally seen persisting for quite a long time at the base of the germling trichome (Text-figs. 7-10).

In the living material which was brought to the laboratory the entire contents of the heterocysts were sometimes observed to come out of the heterocyst wall gradually without undergoing any division (Text-figs. 11, 12 and 13) and were still connected with the rest of the trichome. The contents which have come out of the old wall secrete

a new wall and become a heterocyst again. This evidently represents only a case of rejuvenation of the contents of the heterocyst and not of germination. In one case the contents of the heterocyst have rejuvenated twice. After coming out of the wall a short distance the contents evidently surrounded itself with a new wall and was coming out of the new wall again (Text-fig. 14 ; Pl. I, Fig. 7).

Rivularia Mangini Frey

This alga was collected on moist rocks near a waterfall at Tirupati in South India. This collection which was preserved in 4% formalin on the spot proved interesting since a few germinating heterocysts were found in it. As in the case of *Glæotrichia* described above, a two-celled germling was found enclosed inside the heterocyst wall (Text-fig. 15 ; Pl. I, Fig. 8). The three-celled or four-celled germling



Text-figs. 17-27.—Figs. 17-20. Stages in the germination of the heterocyst in *Calothrix Weberti* Schmidle (after Steinecke). $\times 1900$. Figs. 21-25 and 27. Stages in the germination of the heterocyst in *Nostoc commune* Vaucher (after Geitler). Fig. 26. A two-celled germling in a heterocyst in *Brachytrichia Balani* (Lloyd) Born, and Flah. (after Iyengar and Desikachary).

stages were not found in the material. But a few long germlings were found with traces of the old heterocyst wall still persisting as a cap at the base. These are cases of germination of the heterocyst under natural conditions, since the material was not treated in any way in the laboratory but was preserved as soon as collected. The germination of the heterocyst in *Brachytrichia Balani* recorded by Iyengar and Desikachary (1944, p. 46, Fig. 7 i) was also a case of germination under natural conditions (Text-fig. 26).

DISCUSSION

Geitler (1921), Hollerbach (1928), Canabæus (1929) and Steinecke (1931) have recorded the germination of the heterocysts in several blue-green algæ. In all these cases they found that the contents by a series of transverse divisions became a germling. In most of these cases only a few divisions of the contents were followed. But all the various stages of germination of the contents and the development of the germling into long filaments appears to have been followed so far only in three members of the Cyanophyceæ, viz., *Nostoc commune* (Geitler, 1921), *Tolypothrix Elenkinii* (Hollerbach, 1928) and *Calothrix Weberi* (Steinecke, 1931).

According to Geitler in *Nostoc commune* the contents of the heterocyst first divide to form a two-celled germling (Text-figs. 22 and 23) which by further division becomes a four-celled germling (Text-figs. 24 and 27). This four-celled germling becomes free either through a circumcissal break of the heterocyst wall at the middle into two pieces (Text-figs. 25 and 27) or comes out through one of the pores (Text-figs. 22 and 24). The germling after coming out, by further division, grows into a filament.

The stages of germination of the heterocyst in *Calothrix Weberi* (Text-figs. 17-20) as recorded by Steinecke (1931) are somewhat different from those recorded by Geitler in *Nostoc commune*. The contents of the heterocyst in *Calothrix Weberi* by a transverse division form a two-celled germling (Text-fig. 20). The cell close to the pore of the heterocyst squeezes itself out of the heterocyst wall through the pore (Text-figs. 19 and 20) and then by further divisions gradually grows into a mature filament (Text-fig. 17). The second cell which remains inside degenerates (Text-figs. 17 and 19).

In the two algæ dealt with in this paper not even a single case was observed of the emerging out of the upper cell to form a new filament and the degeneration of the lower cell as recorded by Steinecke (1931) in *Calothrix Weberi*. The writer's observations on the germination of the heterocyst in *Glæotrichia Raciborskii* agree closely with those of Geitler on *Nostoc commune*. But the germling in *Gl. Raciborskii* does not emerge from the heterocyst wall through a break in the middle, but emerges out of the heterocyst wall always through the widening of the pore.

Finally no case of godidia formation by the heterocyst as recorded by Brand and Spratt was observed in the two present algæ.

SUMMARY

Germination of the heterocysts has been known only in five genera, viz., *Nostoc*, *Anabaena*, *Tolypothrix*, *Calothrix* and *Brachytrichia*. The germination of the heterocyst in two more genera, viz., *Glaotrichia* (*Gl. Raciborskii*) and *Rivularia* (*R. Mangini*) is described in the paper.

The details of germination agree closely with those recorded by Geitler in *Nostoc commune*. The stages of germination of the heterocyst described by Steinecke in *Calothrix Weberi*, viz., the emerging out of the upper cell of the two-celled germling and the degeneration of the lower cell, were not observed in the forms investigated here.

The author wishes to express his great indebtedness to Prof. M. O. P. Iyengar, M.A., Ph.D. (Lond.), F.L.S., for his constant guidance and help during the course of this investigation and in the preparation of this paper. His sincere thanks are also due to the authorities of the University of Madras for the award of a research scholarship during the tenure of which the present investigation was carried out.

REFERENCES

- Bharadwaja, Y. (1940) ... "Some aspects of the study of the Myxophyceæ," Presidential address to the Section of Botany, *Proc. Indian Science Congress*, 163.
- Brand, F. (1901) ... "Bemerkungen über Grenzzellen und über spontan rote Inhaltskörper der Cyanophyceen," *Ber. deutsch. Bot. Ges.*, 19 (cited from Geitler, 1932 and 1936).
- Canabœus, L. (1929) ... "Über die Heterocysten und Gasvakuolen der Blaualgen und ihre Beziehungen zueinander," *Pflanzenforschung*, 13, 1.
- Geitler, L. (1921) ... "Versuch einer Lösung des Heterocystenproblems," *Sitzber. Akad. Wiss. Wien math.-nat. Kl.*, Abt. I, 130, 223.
- (1932) ... "Cyanophyceæ" in Rabenhorst's *Kryptogamenflora von Europa*, etc., 14, *Die Algen*, Leipzig.
- (1936) ... "Schizophyceen" in *Handbuch der Pflanzenanatomie*, Abt. II, 6, Teilb. 1b, Berlin.
- Hollerbach, M. M. (1923) ... "De nova specie sectioneque *Tolypothrix* usw.," *Not. Syst. Inst. Crypt. Horti. Bot.*, Petropol, U.S.S.R., 2 (cited from Geitler, 1932 and 1936).
- (1928) ... "Sur la Morphologie de *Tolypothrix Elenkini* Holl., dans les conditions d'habitation naturelle et de laboratoire," *Anch. Russ. Protistol.*, 7, 159. (cited from Geitler, 1932).
- Iyengar, M. O. P., and Desikachary, T. V. (1944) ... "A Systematic Account of Some Marine Myxophyceæ of the South Indian Coast," *Journ. Madras Univ.*, 16, no. 1, Sec. B, 37.
- Spratt, E. R. (1911) ... "Some observations on the life-history of *Anabaena Cycadeæ*," *Ann. of Botany*, 25, 369.
- Steinecke, F. (1931) ... "Das Auskeimen alter Heterocysten bei *Calothrix Weberi*," *Bot. Archiv.*, 34, 153.



1



2



3



4



5



6



7



8

T. V. DESIKACHARY—

*GERMINATION OF THE HETEROCYST IN GLÆOTRICHIA
RACIBORSKII WOŁOSZ. AND RIVULARIA MANGINI FREMY*

EXPLANATION OF THE PLATE

Figs. 1-7. Stages in the germination of the heterocyst in *Glæotrichia Raciborskii* Wolosz.

- Fig. 1. The beginning of the division of the contents of the heterocyst.
- Fig. 2. Two-celled germling.
- Fig. 3. Four-celled germling.
- Fig. 4. A germling with traces of the heterocyst wall still present at the base growing still connected with the old trichome.
- Fig. 5. Three-celled germling.
- Fig. 6. A well-grown germling with the heterocyst wall covering the base.
- Fig. 7. Heterocyst rejuvenating for a second time. Note the two heterocyst walls.
- Fig. 8. A two-celled germling surrounded by a brownish mucilage in *Rivularia Mangini* Frey.

(Figs. 1, 5, 7, $\times 1900$; Figs. 2, 3, $\times 1200$; Fig. 4, $\times 1100$; Figs. 6, 8, $\times 1650$.)